



Course: AI and ML for Robotics

Lab Sheet 4: AIML Techniques for Localization

Objective:

To implement AIML technique to optimize a few components of the traditional localization algorithm.

Below include FOUR DISTINCT use cases, which are designed in this lab sheet to complement existing traditional algorithms.

Tools and Libraries Required:

Programming language: Python

Platform : Google Colab T4 GPU

Simulation : PyBullet

Use case 1:

Camera fed erroneous sensor data is denoised using autoencoder. The output of this is fed to the measurement model before the correction step in localization.

Design:

Input : Noisy Sensor data

Output : Clean Sensor data

Idea : Use denoising Autoencoder

Where does it fit in phases of Robotics?

Before the measurement model is applied in the correction step of localization, the input images of the surroundings can be processed using this to get clean sensor data. Post which optional object(landmark) identification tasks can be done for calculating $P(Z|X)$

Implementation pipeline

<Refer to the Lab sheet for more details>

Use case 2:

In grid based localization, determine the coarseness(Resolution) of the grid based on uncertainties observed in a dynamic environment using Q-Learning.

Design:

Input State : Info abt Tasks Categorization & Uncertainty at timestep(t) Eg.,[Category is HighPrecisionNeededTask , Uncertainty=Low]]



Output Action : Recommendation on necessary resolution/level/No.of.Particles eg.,
[CoarseResolution] or [LessNoofParticles] vs [FineResolution] or
[MoreNoofParticles].

Idea: Use Reinforcement Learner to decide level of redesign(resolution/Particle size required)

Where does it fit in phases of Robotics?

After estimation of pose X_t (Predict & Correct step) check if environment / Robot hardware changes drastically. If so, trigger RL to learn before proceeding with grid use/resampling for estimating the next iteration for predicting $X(t+1)$. Alternatively if its static env, set some frequency eg., once in a week or months for retraining the agent

Implementation pipeline

<Refer to the Lab sheet for more details>

Use case 3:

In Montecarlo localization resampling step, dynamically weigh the particles based on its sensor reliability categorization using Random Forest Classifier.

Design:

Input : Sensor Readings w.r.t each particle Poses ie., Measurement Z_{ei} per particle X_i

Output : adjusted weights per Particles X_i

Idea : Use ensemble learners for prediction of reliability score per particle. This is possible only if supervised input is available

Where does it fit in phases of Robotics?

After estimation of pose X_t in Predict step ie., motion model in the Correct step ie., in Measurement model apply this for particle set (all M).

Implementation pipeline

<Refer to the Lab sheet for more details>

Use case 4:

Dynamic resampling scheduler with particle's measurement observation using Multi Armed Bandit algorithm.

Design:

Input : Localization accuracy observed at previous step ($t-1$), Threshold ESS used in previous step ($t-1$), Bel X_i per particle X_i at current step ($t-1$)



Output : Best ESS Threshold

Idea : Use a Multiarmed bandit with reward = $f(\text{localization accuracy, resampling cost})$ and action space = $\{\text{Threshold1, Threshold2,....}\}$

Where does it fit in phases of Robotics?

After one iteration of localization ie., estimation of pose X_t using aggregation from resampled set M is obtained, call MAB which revises the $Q(\text{ThresholdUsed at } t-1)$. Then recommends the next best ESS threshold to set ie., when to next resampling.

Implementation pipeline

<Refer to the Lab sheet for more details>

Practice Questions:

1. Run the Traditional localization algorithm and collect data. Curate the necessary datasets/reward designs from EDA on the collected data.
 2. Use the data/reward distribution to Integrate respective above solutions separately to train the models
 3. Integrated the trained models into the traditional localization algorithm. Observe the results.
 4. Alternatively, the entire traditional localization pipeline may be replaced by ML/ DNN/ DRL models. The key is in data availability & readiness.
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